

CE 310

Excel Tutorial — Poisson Distribution in Excel

Week 4 · Session 2 · Hands-on Walkthrough

University of Arizona · Dr. Wooyoung Jung · Fall 2026

Today's Plan — Five New Skills, Then the In-Class Exercise

By the end of this walkthrough you will know how to:

1. Use **COUNTIFS with a number criterion** — `G2:G3653, 1` — to count rain-day events by season
2. Estimate λ (average event rate) as events $\div$ periods
3. Compute **$P(X = k)$** with `POISSON.DIST(k,  $\lambda$ , FALSE)` — the PMF
4. Compute **$P(X \leq k)$** with `POISSON.DIST(k,  $\lambda$ , TRUE)` — the CDF
5. Compute **exceedance** $P(X > k)$ and **return period** $P(\geq 1 \text{ in } N \text{ years})$
6. Build a **PMF fill-down table** and insert a **comparison bar chart** (Spring vs Summer)

Assumed from Weeks 1–3: COUNTIFS with text criteria, fill handle, AVERAGEIFS argument order. Today's syntax is nearly identical — only the function name and one argument change.

Step 1 — Open the Dataset

File: `Week4_Tucson_Precipitation_Daily_2013_2022.csv` · 3,652 rows · 1 row per day

Column	Header	Type
A	Date	Date
B	Year	Number (2013–2022)
C	Month	Number (1–12)
D	Day	Number (1–31)

💡 **IsRainDay** is the Poisson event indicator.
Counting IsRainDay = 1 per month gives k .
Averaging those k values across months gives λ .

Save as `.xlsx` immediately. Verify:
`=COUNTA(A:A)-1` → **3,652**.

Step 1 (cont.) – Dataset Columns

Column	Header	Type
E	Season	Text ("Winter/Spring/Summer/Fall")
F	PRCP_in	Number (inches)
G	IsRainDay	Number (1 = rain; 0 = dry)

Step 2 — COUNTIFS with a Number Criterion

In Week 3 criteria were text strings. For IsRainDay and Month, the criterion is a **number** — no quotes needed.

 **Type these and confirm:**

=COUNTIFS(G2:G3653, 1) → all rain days → **463**

=COUNTIFS(E2:E3653, "Summer", G2:G3653, 1) → Summer rain days → **197**

=COUNTIFS(E2:E3653, "Spring", G2:G3653, 1) → Spring rain days → **42**

=COUNTIFS(C2:C3653, 7, G2:G3653, 1) → July rain days (Month = 7) → **103**

G2:G3653, 1 matches the number 1 — **no quotes**. C2:C3653, 7 matches July — **no quotes**.
Quotes are only for text like "Summer".

 Writing "1" (with quotes) instead of 1 compares text to numbers — Excel returns **0 with no error**.
If your count is zero when you expect hundreds, check for accidental quotes around the number.

Step 2 — Quotes Or No Quotes

Season is text. IsRainDay is a number.

	E	F	G
1	Season	PRCP_in	IsRainDay
9	Winter	0.00	0
10	Winter	0.00	0
11	Winter	0.00	0
12	Winter	0.03	1
:			
3653			

Two criteria, two rules

=COUNTIFS(E2:E3653,"Summer")

text — quotes required

=COUNTIFS(G2:G3653,1)

number — no quotes

=COUNTIFS(G2:G3653,"1")

text "1" against a number column

=COUNTIFS(G2:G3653, 1) counts 463 rain days across the ten years.

X The quoted version returns 0 — and no error message.

Step 3 — λ from Data: Events ÷ Periods

$$\hat{\lambda} = \frac{\text{total events observed}}{\text{total observation periods}}$$

Compute these in blank cells and keep them — the in-class exercise will reference them:

Calculation	Formula	λ value
Overall (120 months)	=463/120	3.858 rain days/month
Summer (30 months)	=197/30	6.57 rain days/month
Spring (30 months)	=42/30	1.40 rain days/month
July (10 Julys)	=103/10	10.3 rain days/July

Step 3 — Four λ Values, One Rule

λ is a count divided by the periods it was counted over

Ten years of daily records. The period you divide by is the period λ then describes.

All months	$\frac{463}{120}$	rain days ÷ months	=	3.858	rain days per month
Summer	$\frac{197}{30}$	rain days ÷ summer months	=	6.57	rain days per month
Spring	$\frac{42}{30}$	rain days ÷ spring months	=	1.40	rain days per month
July only	$\frac{103}{10}$	rain days ÷ Julys	=	10.3	rain days per July

Step 4 — POISSON.DIST with FALSE: The PMF

`=POISSON.DIST(k, λ, FALSE)` → $P(X = k)$ — probability of **exactly** k events.

The third argument controls what the function returns: `FALSE` = one probability (PMF), `TRUE` = running total up to k (CDF).

 **Type these and confirm — multiply by 100 for a percentage:**

`=POISSON.DIST(0, 3.858, FALSE)*100` → $P(\text{dry month} \mid \text{overall}) \approx \mathbf{2.11\%}$

`=POISSON.DIST(5, 3.858, FALSE)*100` → $P(\text{exactly 5 rain days}) \approx \mathbf{15.0\%}$

`=POISSON.DIST(0, 10.3, FALSE)*100` → $P(\text{dry July} \mid \lambda = 10.3) \approx \mathbf{< 0.01\%}$

 **POISSON.DIST returns a decimal (0 to 1) — not a percentage. Always multiply by 100 before entering the answer sheet. The formula returns 0.0208, not 2.08.**

Step 5 — POISSON.DIST with TRUE: The CDF

`=POISSON.DIST(k, λ, TRUE)` → $P(X \leq k)$ — probability of **k or fewer** events.

Only the last argument changes: `FALSE` → `TRUE`. The CDF adds up all PMF values from $P(X=0)$ through $P(X=k)$.

Compare PMF vs CDF at the same k:

PMF: `=POISSON.DIST(5, 3.858, FALSE)*100` → $\approx 15.0\%$ (exactly 5 rain days)

CDF: `=POISSON.DIST(5, 3.858, TRUE)*100` → $\approx 80.7\%$ (5 or fewer rain days)

At $k=0$: PMF and CDF are equal — $P(X=0)$ is the only term in the sum.

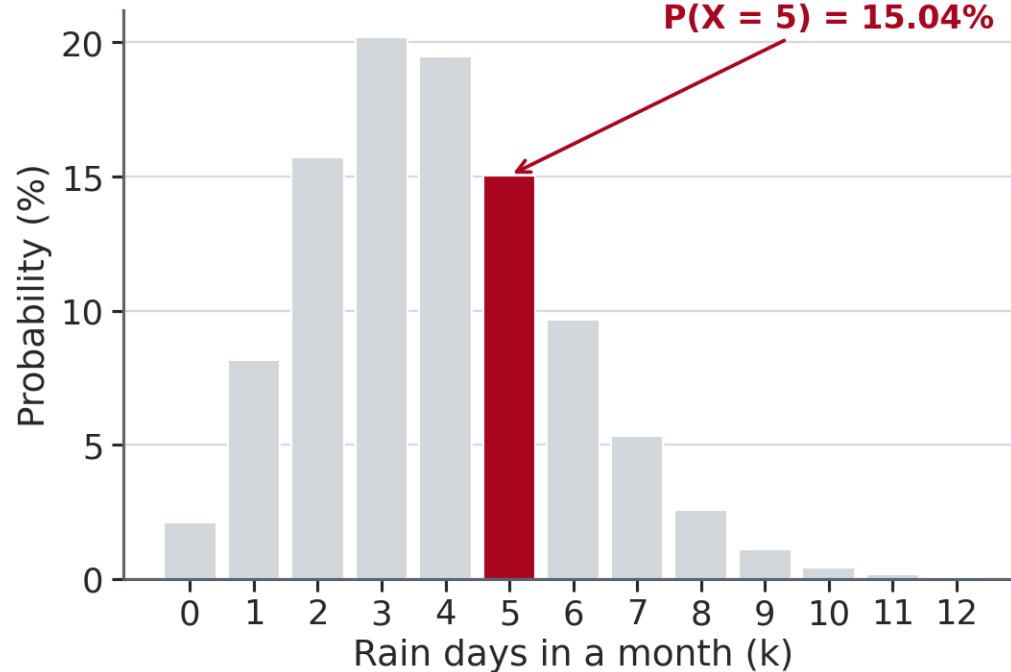
 The CDF (80.7%) means roughly 80% of months have 5 or fewer rain days. The other 20% exceed 5 — those are the months that stress infrastructure sized for the average.

Steps 4–5 — FALSE and TRUE, Drawn

POISSON.DIST(k, 3.858, ...) — the third argument is the whole difference

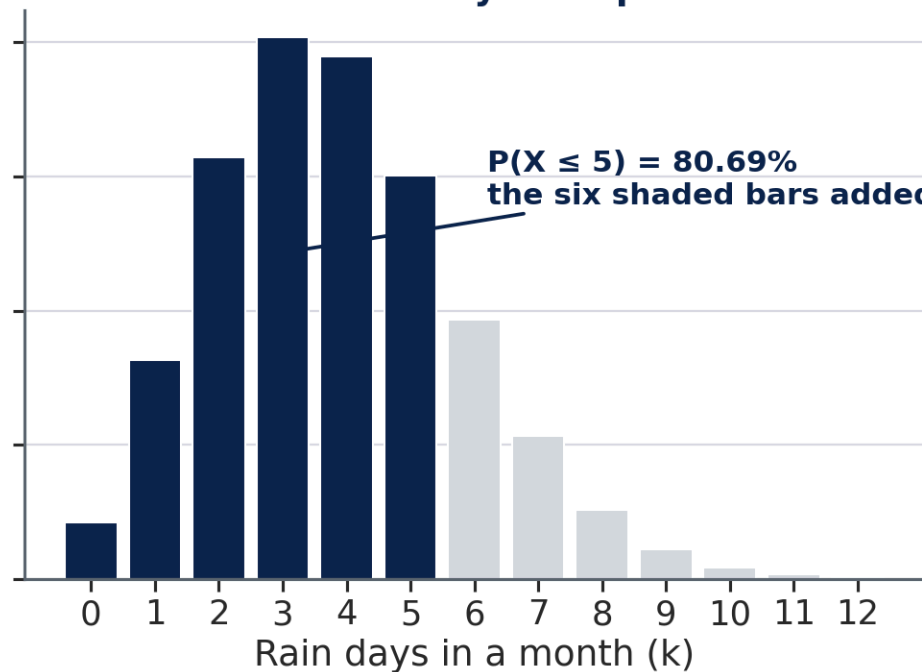
FALSE — one bar

$P(X = 5) = 15.04\%$



TRUE — every bar up to k

$P(X \leq 5) = 80.69\%$
the six shaded bars added up



Step 6 — Exceedance: $P(X > k) = 1 - \text{CDF}$

Exceedance probability answers: how often does demand exceed the design threshold?

$$P(X > k) = 1 - \text{POISSON.DIST}(k, \lambda, \text{TRUE})$$

 **Type these and confirm:**

`=(1 - POISSON.DIST(5, 3.858, TRUE))*100` → $P(X > 5 \mid \text{overall}) \approx \mathbf{19.3\%}$

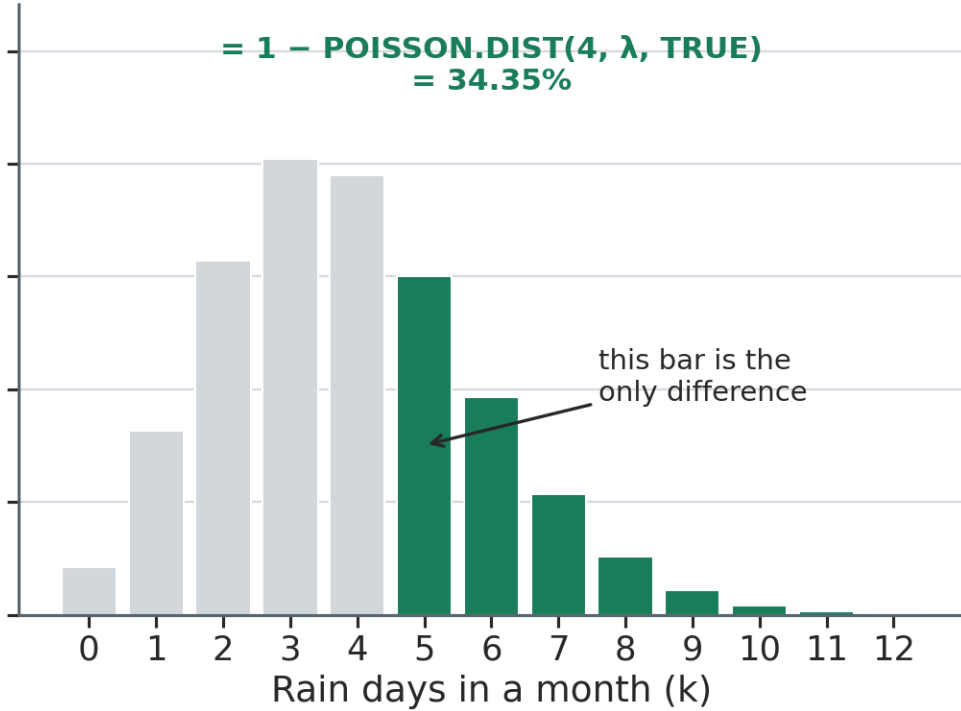
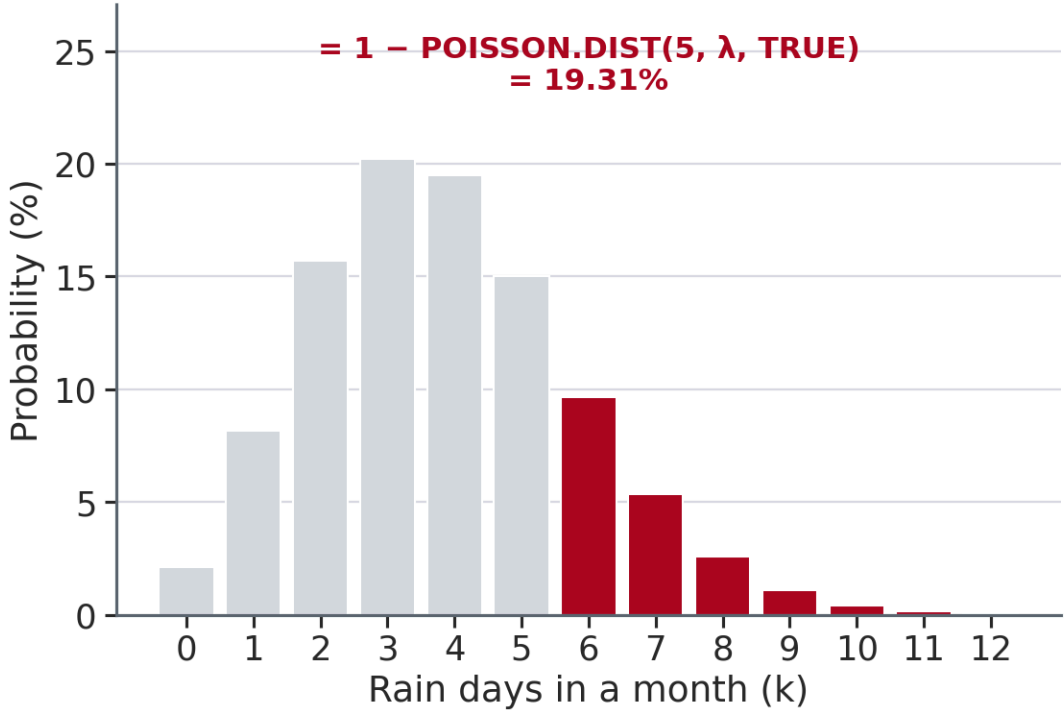
`=(1 - POISSON.DIST(4, 1.40, TRUE))*100` → $P(X \geq 5 \mid \text{Spring}) \approx \mathbf{1.43\%}$

The second formula uses `k-1 = 4`. The logic: "at least 5" means its complement is "4 or fewer," so $P(X \geq 5) = 1 - P(X \leq 4) = 1 - \text{CDF}(4)$. Use `k-1` in the CDF, not `k`.

Question	Uses	Formula
P(more than k events)	<code>POISSON.DIST(k, λ, TRUE)</code>	<code>=1 - POISSON.DIST(k, λ, TRUE)</code>
P(at least k events)	<code>POISSON.DIST(k-1, λ, TRUE)</code>	<code>=1 - POISSON.DIST(k-1, λ, TRUE)</code>

Step 6 – Which Bars Are You Adding Up?

“At least k” passes k-1 to the CDF — and the gap is exactly $P(X = 5) = 15.04\%$
more than 5 ($X > 5$) at least 5 ($X \geq 5$)



Step 7 — Return Period: $=1 - \text{EXP}(-N/T)$

A **T-year event** has annual rate $\lambda = 1/T$. Over N years, $P(\text{at least one event}) = 1 - e^{(-N/T)}$.

In Excel: `= (1 - EXP(-N/T)) * 100`

 **25-year storm, 10-year project life:**

`= (1 - EXP(-10/25)) * 100` → $\approx$ **32.97%** — nearly one in three projects

Scenario	N	T	Result
25-year storm, 10-year project	10	25	33.0%
100-year storm, 50-year project	50	100	39.3%

 Use `EXP()` — not `POISSON.DIST()` — for return period calculations. Both give the same answer, but the exponential form $1 - e^{(-N/T)}$ is standard engineering practice — it connects directly to the Poisson zero-event probability.

Step 8 — PMF Fill-Down Table (Chart 1 Setup)

Build a $k = 0$ to 15 table comparing Spring ($\lambda = 1.40$) vs Summer ($\lambda = 6.57$). In the in-class exercise answer sheet, Spring λ (question B2) is in cell E21 and Summer λ (question A7) is in cell E15:

 **In a blank area of the answer sheet:**

I1: k J1: Spring PMF K1: Summer PMF

I2:I17 → type 0, 1, 2, ..., 15 (type 0 and 1, select both, drag down)

J2: =POISSON.DIST(I2, \$E\$21, FALSE) ← Spring λ (answer B2)

K2: =POISSON.DIST(I2, \$E\$15, FALSE) ← Summer λ (answer A7)

Select J2:K2 → double-click fill handle → fills to row 17.

Why \$E\$21 and \$E\$15 ? The \$ signs lock the λ cells so they don't shift as the formula fills down. The k cell (I2) has no \$ — it advances row by row.

Without \$, each row looks at a different λ cell and the probabilities come out wrong.

Step 9 — Insert the PMF Comparison Chart

Select **I1:K17** — the k, Spring PMF, and Summer PMF columns including headers.

 **Insert** → **Charts** → **Insert Column or Bar Chart** → **Clustered Column**

Spring and Summer bars appear side by side for each k. Then format:

- Double-click the title → rename: **"PMF: Spring vs Summer Rain Days per Month"**
- Chart Design → Add Chart Element → Axis Titles
Y axis: **"Probability"** · X axis: **"Rain days in month (k)"**
- Right-click the Y axis → Format Axis → Minimum: **0**
- Chart Design → Add Chart Element → **Legend** — Excel usually adds one as soon as there are two series, so check before adding a second

 Use a **Clustered Column** chart — not a Histogram. Side-by-side bars at the same k let you compare Spring and Summer directly. Leave the gap between clusters; touching bars signal a continuous distribution (histogram), which is wrong for a discrete PMF.

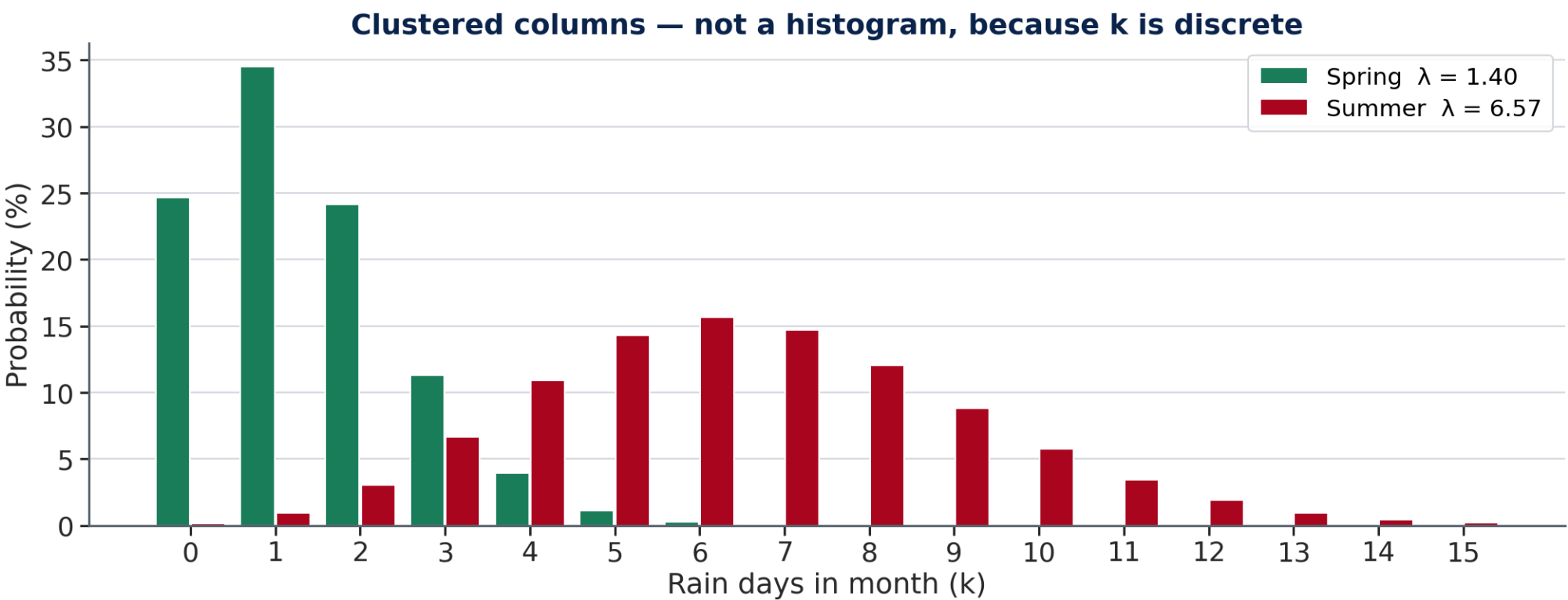
Step 10 — Data Labels for Chart 2

Chart 2 asks for the **λ value printed on each bar** — four seasons, four numbers. That is a data label.

 **Chart Design → Add Chart Element → Data Labels → Outside End**

Same menu as the axis titles you just used. Add them **only when a week asks:** on the 16-bar PMF chart above they would be unreadable.

Step 9 — What Your Chart Should Look Like



Spring peaks at one rain day a month; summer at six. Same function, same table, one number different.

Quick Check — Before You Start the In-Class Exercise

Think, then check

You computed $P(\text{dry month} \mid \lambda = 3.858) \approx 2.11\%$ and $P(X \leq 5 \mid \lambda = 3.858) \approx 80.7\%$.

Question A: Write the Excel formula for $P(\text{at least 5 rain days in a spring month} \mid \lambda = 1.40)$. Does the POISSON.DIST use $k = 5$ or $k = 4$?

Question B: The return period formula $=1 - \text{EXP}(-N/T)$ computes $P(\text{at least one event in } N \text{ years})$. What does $\text{EXP}(-N/T)$ equal in Poisson terms?

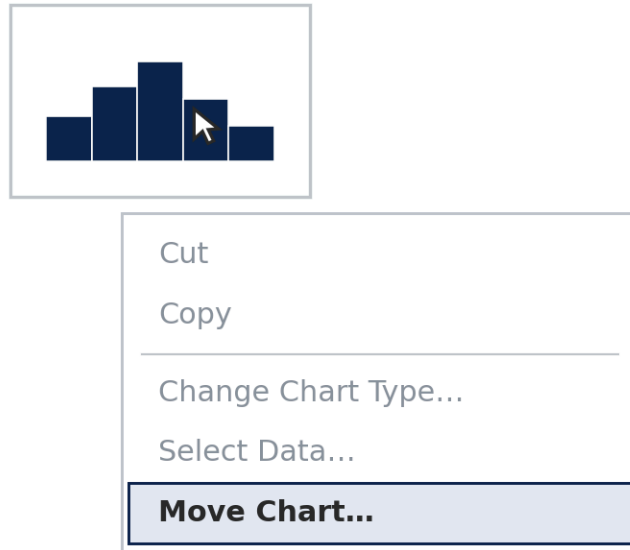
Compare with your neighbor. We'll confirm before opening the in-class exercise.

The Step Every In-Class Exercise Ends With

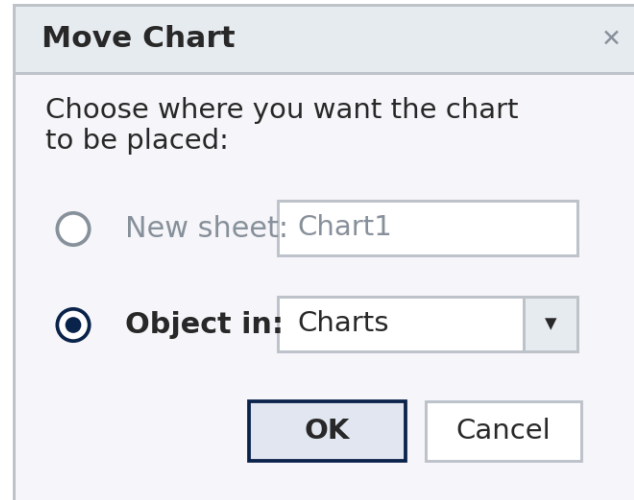
Getting a finished chart into the box it is graded in

Every week. Build the chart on the sheet its data lives on, then move it onto the Charts sheet.

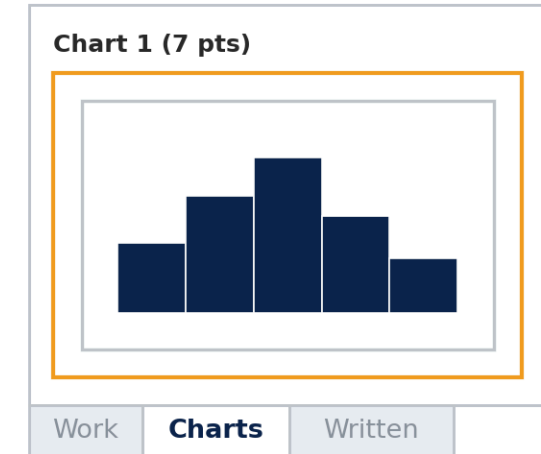
1 Right-click the chart



2 Choose "Object in"



3 It lands in the box



✗ New sheet — makes a separate sheet called Chart1. The box on Charts stays empty and the points are lost.

✓ Object in → Charts — then drag the chart inside the outlined box.

⚠ Leave it a live chart — a pasted picture of one scores zero.

Quick Reference — Week 4 New Formulas

Task	Formula
Season events	<code>=COUNTIFS(E2:E3653,"Summer",G2:G3653,1)</code>
Estimate λ	<code>= events / periods</code>
PMF — $P(X = k)$	<code>=POISSON.DIST(k, λ, FALSE)</code>
CDF — $P(X \leq k)$	<code>=POISSON.DIST(k, λ, TRUE)</code>

Task	Formula
$P(X > k)$ exceedance	<code>=1 - POISSON.DIST(k, λ, TRUE)</code>
$P(X \geq k)$ at-least-k	<code>=1 - POISSON.DIST(k-1, λ, TRUE)</code>
Return period	<code>=1 - EXP(-N/T)</code>
PMF fill-down	<code>=POISSON.DIST(k_cell, \$lambda\$, FALSE)</code>

Before You Submit

⚠ Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

Two most common formula mistakes: (1) number in quotes "1" instead of 1 — returns 0 silently, (2) missing \$ in PMF fill-down — λ cell shifts with each row.

1. **File format** — **.xlsx only**. Save As → Excel Workbook. .xls, .csv, and .pdf cannot be graded.
2. **Sheet name** — **Answers**. Do not rename or delete this sheet.
3. **Column layout** — **do not change**. Answers go in Column E. Do not insert, delete, or move columns.
4. **File name** — **CE310_W4_<NetID>.xlsx** (replace <NetID> with your UA NetID).
5. **Upload** the **.xlsx** file to D2L Dropbox by **9:00 AM next Wednesday**.

You're Ready — Open the In-Class Exercise Instructions

Wk04.03_In_Class_Exercise.pdf · Wk04.04_In_Class_Answer_Sheet.xlsx

Setup steps before Section A:

- Save as .xlsx · Verify =COUNTA(A:A)-1 → **3,652** rows
- Confirm 7 column headers: Date, Year, Month, Day, Season, PRCP_in, IsRainDay

Take-home: B7–B14 (exceedance, winter season, return period), Chart 2, Memo PDF.

Raise your hand if:

- COUNTIFS returns 0 when you expect hundreds → check for quoted numbers ("1" not 1)
- POISSON.DIST returns a tiny decimal → remember * 100 for percentages
- PMF fill-down gives near-zero for all rows → missing \$ on the λ cell reference